

Anaconda vs VS Code Built-in Python

****Prepared by:**** Musawenkosi Bongumusa (Musa) Vilakazi

****Position:**** Senior Risk Analytics Engineer

****Company:**** Standard Bank Group

****Date:**** 08 June 2026

Why Anaconda + VS Code is the Recommended Setup

Comparison Table

Aspect System Python**	**Anaconda (Recommended)**	**VS Code Built-in /
-----	-----	-----

Package Management conflicts	Excellent (conda + pip)	Only pip — frequent
Environment Isolation error-prone	Very easy (`conda create -n risk_model`)	Manual and
Complex Packages (XGBoost, SHAP, etc.) installation errors	Reliable & stable	Often causes
Reproducibility	High (`environment.yml`)	Low
Stability for Banking Work riskier	Preferred in financial institutions	Acceptable but

| Long-term Maintenance | Easy to share and reproduce | More difficult
|

Key Benefits of Using Anaconda

1. ****Isolated Environments**** — Prevents package version conflicts between different projects.
2. ****Reliable Installation**** of data science and machine learning libraries.
3. ****Better for Regulatory & Production Work**** — Easier to maintain consistent environments.
4. ****Export Full Environment**** for sharing:

```
``bash
```

```
conda env export > environment.yml
```